The Wave Equation: Travelling Signals, Energy, and Reflection

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The one-dimensional wave equation can be understood almost entirely through two directions of transport. Its characteristic variables move at speeds c and $-c$, so initial disturbances influence only a finite region of space. The same geometry explains d'Alembert's formula, the domain of dependence, reflection at a fixed boundary and the conversion between standing and travelling waves. Energy conservation makes this propagation picture a uniqueness argument rather than merely a method of guessing solutions.[1]

§1.1 Factoring the Wave Operator

For a vibrating string, $u(x,t)$ denotes its transverse displacement. In the linear small-amplitude model,

$$\boxed{u_{tt} = c^2 u_{xx}, \qquad c > 0.}$$

The spatial curvature determines acceleration rather than the rate of change of displacement. Because the constant-coefficient operators commute,

$$\partial_t^2 - c^2\partial_x^2 = (\partial_t - c\partial_x)(\partial_t + c\partial_x).$$

Thus the wave equation is the composition of the two opposite transport operators.

Introduce characteristic coordinates

$$\xi = x + ct, \qquad \eta = x - ct.$$

Then

$$\partial_t = c(\partial_\xi - \partial_\eta), \qquad \partial_x = \partial_\xi + \partial_\eta,$$

and the wave equation becomes $u_{\xi\eta} = 0$. Integrating twice shows that every sufficiently smooth solution on a characteristic rectangle can be written as

$$\boxed{u(x,t) = F(x - ct) + G(x + ct).}$$

The F-component travels right, while the G-component travels left. The formula is an exact consequence of factorization, not a superposition guess.

§1.2 Displacement and Initial Velocity

On the whole real line, prescribe

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x).$$

At $t = 0$, the travelling-wave representation gives

$$F(x) + G(x) = f(x), \qquad -cF'(x) + cG'(x) = g(x).$$

[1] MIT 18.152 (PDE lecture notes); MIT 2.062J, *Wave Propagation*, Chapter 2.

Differentiate the first equation and solve for

$$F' = \frac{1}{2}f' - \frac{g}{2c}, \qquad G' = \frac{1}{2}f' + \frac{g}{2c}.$$

Substitute these into the travelling-wave formula to obtain d'Alembert's solution:

$$\boxed{u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds.}$$

The first term splits the initial displacement into two travelling profiles; the second accumulates the initial velocity over an expanding interval.

If $f(x) = e^{-x^2}$ and $g = 0$, the initial Gaussian pulse splits into two Gaussian pulses:

$$u(x,t) = \tfrac{1}{2}e^{-(x-ct)^2} + \tfrac{1}{2}e^{-(x+ct)^2}.$$

Unlike heat diffusion, the widths of the travelling components do not grow in this ideal nondispersive model.

§1.3 Finite Propagation Speed

The value $u(x,t)$ in d'Alembert's formula depends only on f at $x \pm ct$ and on g inside

$$[x - ct, x + ct].$$

If the initial data vanish on this interval, then $u(x,t) = 0$. In particular, initial data supported in $[-R, R]$ produce a solution supported within

$$[-R - ct, R + ct]$$

at time $t \geq 0$.

The interval $[x - ct, x + ct]$ is the backward domain of dependence of the event (x,t). In the space-time plane its boundary consists of characteristic lines of slopes $\pm c$. The sharp signal front is a consequence of the hyperbolic operator's factorization; the heat kernel, by contrast, is positive at every point for any positive time.

§1.4 Uniqueness by Energy Conservation

On an interval $[a,b]$, define the local wave energy density and flux

$$e = \frac{1}{2}(u_t^2 + c^2 u_x^2), \qquad j = -c^2 u_t u_x.$$

Differentiate:

$$\begin{aligned} e_t + j_x &= u_t u_{tt} + c^2 u_x u_{xt} - c^2(u_{xt}u_x + u_t u_{xx}) \\ &= u_t(u_{tt} - c^2 u_{xx}) = 0. \end{aligned}$$

Therefore

$$\boxed{e_t + j_x = 0,}$$

an exact local conservation law. Integrating gives

$$\frac{d}{dt}\int_a^b e(x,t)\,dx = j(a,t) - j(b,t).$$

Energy changes within a fixed segment only through its boundaries.

For a finite string $0 < x < L$ with fixed endpoints $u(0,t) = u(L,t) = 0$, differentiating the boundary condition gives $u_t = 0$ there, so the boundary flux vanishes. Hence

$$\boxed{E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)\,dx = E(0).}$$

If two solutions have the same initial and boundary data, their difference has zero initial energy. Since its energy stays zero, both its derivatives vanish, and the zero boundary value forces the difference itself to vanish. Thus the solution is unique.

This energy proof works without explicitly constructing the travelling waves. It also suggests how to handle variable coefficients and higher-dimensional problems where d'Alembert's elementary formula is unavailable.

§1.5 Forcing contributes energy at a calculable rate

Consider the forced equation

$$u_{tt} - c^2 u_{xx} = h(x,t)$$

on the real line with initial data f, g. Duhamel's principle gives

$$\boxed{\begin{aligned} u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds \\ + \frac{1}{2c}\int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} h(y,\tau)\,dy\,d\tau. \end{aligned}}$$

The forcing at time τ affects the event (x,t) only if its spatial position y lies within the backward characteristic cone. This makes finite propagation explicit for the forcing as well as for the initial data.

To check the formula, set $f = g = 0$ and $h \equiv 1$. The inner integral is $2c(t-\tau)$, so

$$u(x,t) = \int_0^t (t-\tau)\,d\tau = \frac{t^2}{2}.$$

Indeed $u_{tt} = 1$ and $u_{xx} = 0$. In the energy identity, forcing adds the source term

$$e_t + j_x = h\,u_t,$$

which is the local rate of work done by the external force.

§1.6 Wave Reflection at a Fixed End

On the half-line $x > 0$ with fixed endpoint $u(0,t) = 0$, extend the initial displacement and velocity to the whole line as odd functions:

$$\widetilde{f}(-x) = -\widetilde{f}(x), \qquad \widetilde{g}(-x) = -\widetilde{g}(x).$$

The whole-line d'Alembert solution from these odd data remains odd in x, hence vanishes at $x = 0$. Restricting it to $x > 0$ solves the fixed-end problem.

This is the method of images. A right-moving incoming pulse is reflected into a left-moving pulse with reversed sign so that the endpoint displacement stays zero. For a free endpoint with Neumann condition $u_x(0,t) = 0$, the analogous construction uses even extensions, giving reflection without sign reversal.

The boundary condition therefore selects which symmetry extension makes the whole-line solution compatible with the physical endpoint.

§1.7 Fourier modes are reflected travelling waves

On $0 < x < L$, the fixed endpoint conditions select the sine eigenfunctions

$$X_n(x) = \sin \frac{n\pi x}{L}, \qquad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

For sufficiently regular data satisfying the compatibility conditions, the solution is

$$\boxed{u(x,t) = \sum_{n\geq 1} \left(a_n \cos \frac{n\pi ct}{L} + b_n \sin \frac{n\pi ct}{L}\right) \sin \frac{n\pi x}{L},}$$

with coefficients

$$a_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \qquad b_n = \frac{2}{n\pi c}\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.$$

Each mode is an independent harmonic oscillator in time. Its fixed spatial nodes are caused by interference between oppositely travelling components:

$$\sin(kx)\cos(kct) = \tfrac{1}{2}\big(\sin(k(x+ct)) + \sin(k(x-ct))\big).$$

A standing wave is therefore not a third kind of propagation. It is an exact superposition of two travelling waves of the same frequency, constrained by reflections at the endpoints.

§1.8 Damping shows exactly what energy conservation uses

If the string experiences linear damping,

$$u_{tt} + \gamma u_t = c^2 u_{xx}, \qquad \gamma > 0,$$

the same multiplication by u_t gives, under fixed endpoints,

$$\boxed{\frac{dE}{dt} = -\gamma \int_0^L u_t^2\, dx \leq 0.}$$

The boundary still contributes no energy, but damping removes kinetic energy in the interior.

The contrast with the heat equation is now precise. Both equations admit Fourier decompositions, yet a heat mode has multiplier $e^{-\kappa\lambda_n t}$, while a wave mode oscillates at frequency $c\sqrt{\lambda_n}$. The two transport directions of the wave equation combine into reversible oscillatory motion; heat flow defines a dissipative semigroup with no time-reversed smoothing operation. The geometry of characteristics and the energy identity both reveal this distinction.